

# HANDOUT TITLE

SPRING 2026 · COURSE NAME

## 1 Energy and Conservation

A particle of mass  $m$  moves in a one-dimensional potential  $V(x)$ . The total mechanical energy is

$$E = \frac{1}{2}mv^2 + V(x), \quad (1)$$

which is conserved when no external forces act.

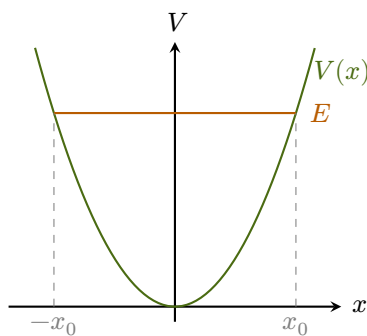
**Definition 1.1.** A force is *conservative* if it can be written as the negative gradient of a potential,  $F(x) = -dV/dx$ .

**Theorem 1.2.** *For a conservative force, the total mechanical energy  $E$  is constant along any solution of the equation of motion.*

*Remark 1.3.* The converse fails in general: dissipative systems admit no such potential, and  $E$  is not conserved.

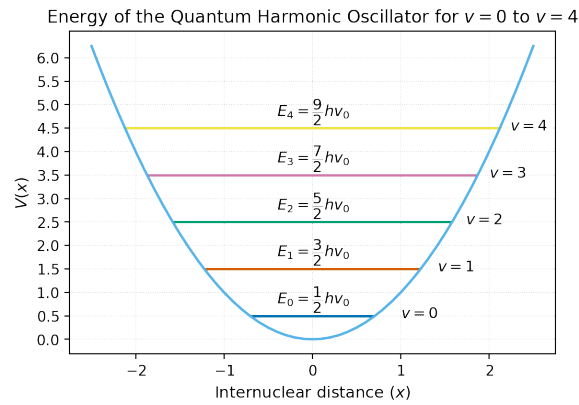
### 1.1 Potential Energy Diagrams

A plot of  $V(x)$  against  $x$  summarizes the qualitative motion at a glance: the particle is confined to regions where  $E \geq V(x)$ , and turning points occur where  $E = V(x)$ , as illustrated in Figure 1.



**Figure 1.** Harmonic potential (olive) with energy level  $E$  (orange) and classical turning points  $\pm x_0$  (gray).

The subsection heading above sits one size step below the section title, while remaining at body size — enough to mark a finer level when the handout is used for lecture notes.



**Figure 2.** Schematic figure for the topic.

### Exercise 1.1 - Simple harmonic oscillator

Consider a particle of mass  $m$  in the potential  $V(x) = \frac{1}{2} k x^2$ .

- Write down the equation of motion using Newton's second law.
- Show that  $E = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$  is conserved along solutions.

The result can be verified by differentiating  $E$  with respect to time and using the equation of motion from part (a).

- Find the angular frequency of small oscillations.

### Exercise 1.2 - Inclined plane

A block of mass  $m$  slides down a frictionless inclined plane that makes an angle  $\theta$  with the horizontal.

- Find the acceleration along the plane.
- How long does it take to slide a distance  $L$  from rest?
- What is the speed at the bottom?

## 2 Circular Motion

In uniform circular motion, the centripetal acceleration has magnitude  $a_c = v^2/r$ , directed toward the center of the circle.

### Exercise 2.1 - Circular orbit

A satellite moves in a circular orbit at radius  $r$  around a planet of mass  $M$ .

- a) Show that the orbital speed is  $v = \sqrt{GM/r}$ , where  $G$  is Newton's gravitational constant.
- b) Express the orbital period  $T$  in terms of  $r$ ,  $M$ , and  $G$ .

### 3 Numerical Example

The `python` environment typesets code with Gruvbox Light syntax highlighting. The short script below verifies the orbital speed and period from Exercise 2.1 numerically for three representative radii.

```
1 import numpy as np
2
3 G = 6.674e-11      # gravitational constant (m^3 kg^-1 s^-2)
4 M = 5.972e24       # Earth mass (kg)
5
6 orbits = {"Low orbit": 6.40e6,
7           "ISS (approx)": 6.78e6,
8           "GEO": 4.22e7}
9
10 for name, r in orbits.items():
11     v = np.sqrt(G * M / r)          # orbital speed [m/s]
12     T = 2 * np.pi * r / v / 3600  # orbital period [hours]
13     print(f"{name:14s} v = {v/1e3:.3f} km/s T = {T:.2f} h")
```